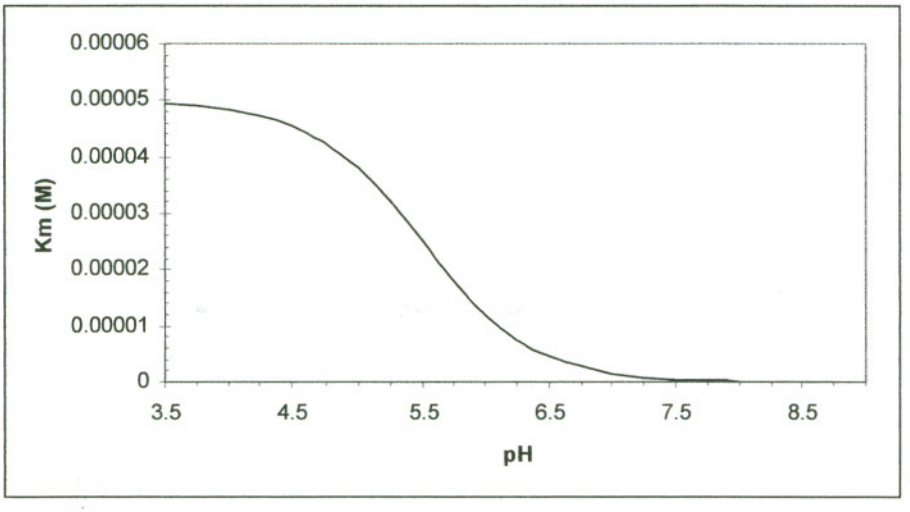
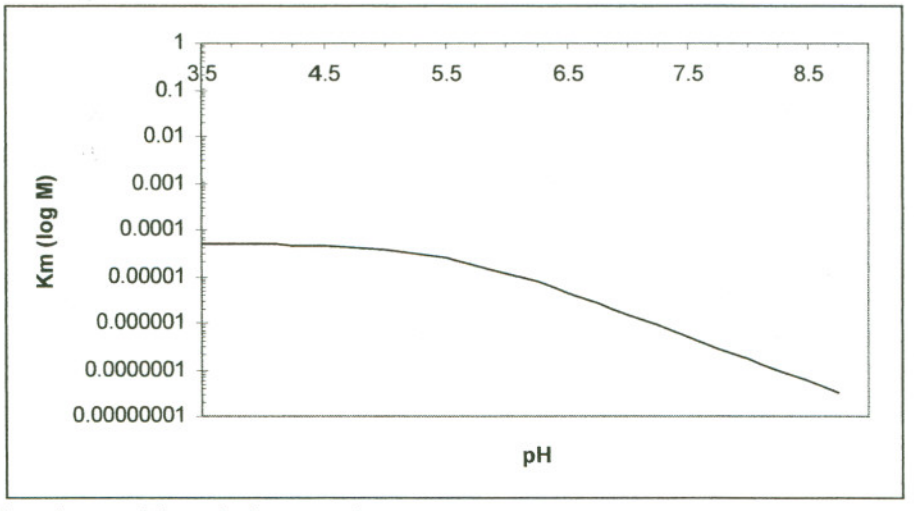
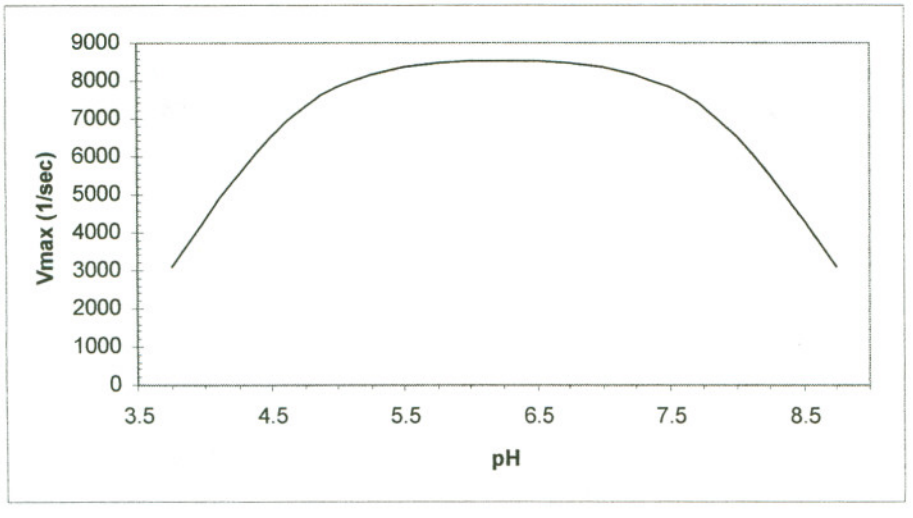
Biochemistry Cumulative Examination

January 21, 2005

Suppose you are studying the activity of a new enzyme. Other experiments have proven that a glutamic acid, a histidine, and a lysine are important in the mechanism. **50 points each** for parts **1** and **2**.

1. Consider the two graphs at right (upper and lower), showing substrate binding affinity as a function of pH. Based on these data and what you know about the pK_a for each of the three side chains, which of the three side chain(s) do you believe play a role in binding? What is the protonation state(s) of these/this residue(s) in the high affinity state? Explain your reasoning in as much detail as time and knowledge permit.



2. Consider now the graph below, showing turnover rate as a function of pH. Based on these data and what you know about the pK_a for each of the three side chains, which of the three side chain(s) do you believe play a role in the enzymatic mechanism? What is the protonation state(s) of these/this residue(s) in the most active state? Explain your reasoning in as much detail as time and knowledge permit.